UE26MA141A Engineering Mathematics – I · Unit 1 · Partial Differentiation

Flowcharts and visuals

Procedures, decisions, concept maps and comparison tables.

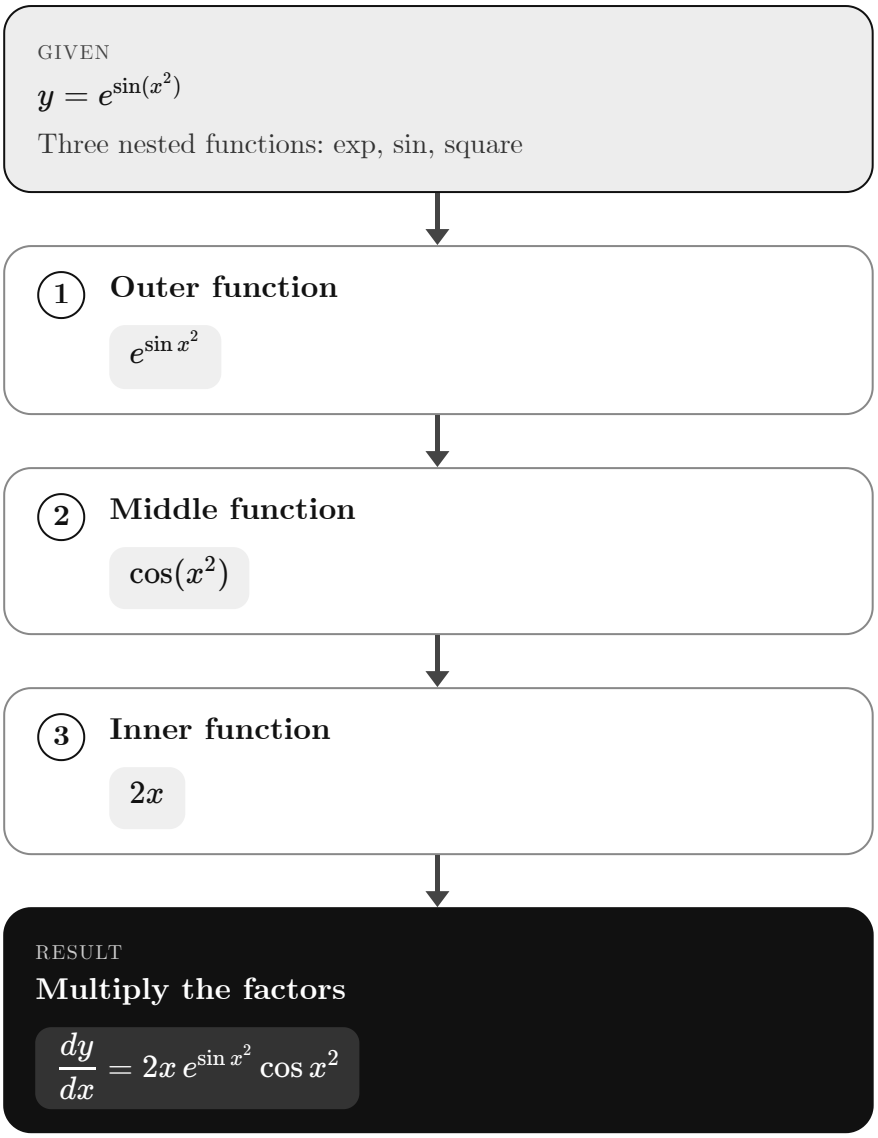
Classes 003–004 · Total derivative, chain rule and implicit functions · compiled 14 October 2026

ESA ★★★	Asked repeatedly in past ESA papers
ESA ★★	Asked in past ESA papers or computed in class-work problems
ESA ★	Conceptual: taught in class, not examined so far
ADDED FOR CLARITY	Standard context that is not in the class material
TRANSCRIPT ONLY	Said in the lecture but not on any slide
VIDEO ONLY	Seen only in a frame of the class video
NOT ON CURRENT SLIDES	From older material or past papers, not this year's slides
S003·7 · S004·9	Deck 003 / 004, slide number (rendered .pptx)
T003a 00:06:27	Transcript 003 part I: “Total derivative, chain rule and implicit functions”
T003b 00:01:10	Transcript 003 part II: “Problems on total derivatives, chain rule”
T004a 00:13:09	Transcript 004 part I: “Implicit functions and problems”
T004b 00:02:31	Transcript 004 part II: “Problems on composite and implicit functions”
CW U1 P14 · ESA Jan-24 s3 1b	Unit 1 class-work problem 14 · past-paper question

Contents

1. Figure F1.1 · Differentiating a composite function from the outside in.
2. Figure F2.1 · Total derivative in three moves.
3. Figure F3.1 · Estimating a value near a known point.
4. Figure F4.1 · Solving a rates-of-change problem.
5. Figure F5.1 · Which rule? Read the chain first.
6. Figure F5.2 · Proving an identity such as $6v_x + 4v_y + 3v_z = 0$.
7. Figure F6.1 · Finding dy/dx for an implicit function: two routes.
8. Figure F7.1 · Partial derivatives from $f(x, y, z) = 0$.
9. Total derivative vs chain rule vs implicit differentiation

1 Composite functions and the one-variable chain rule

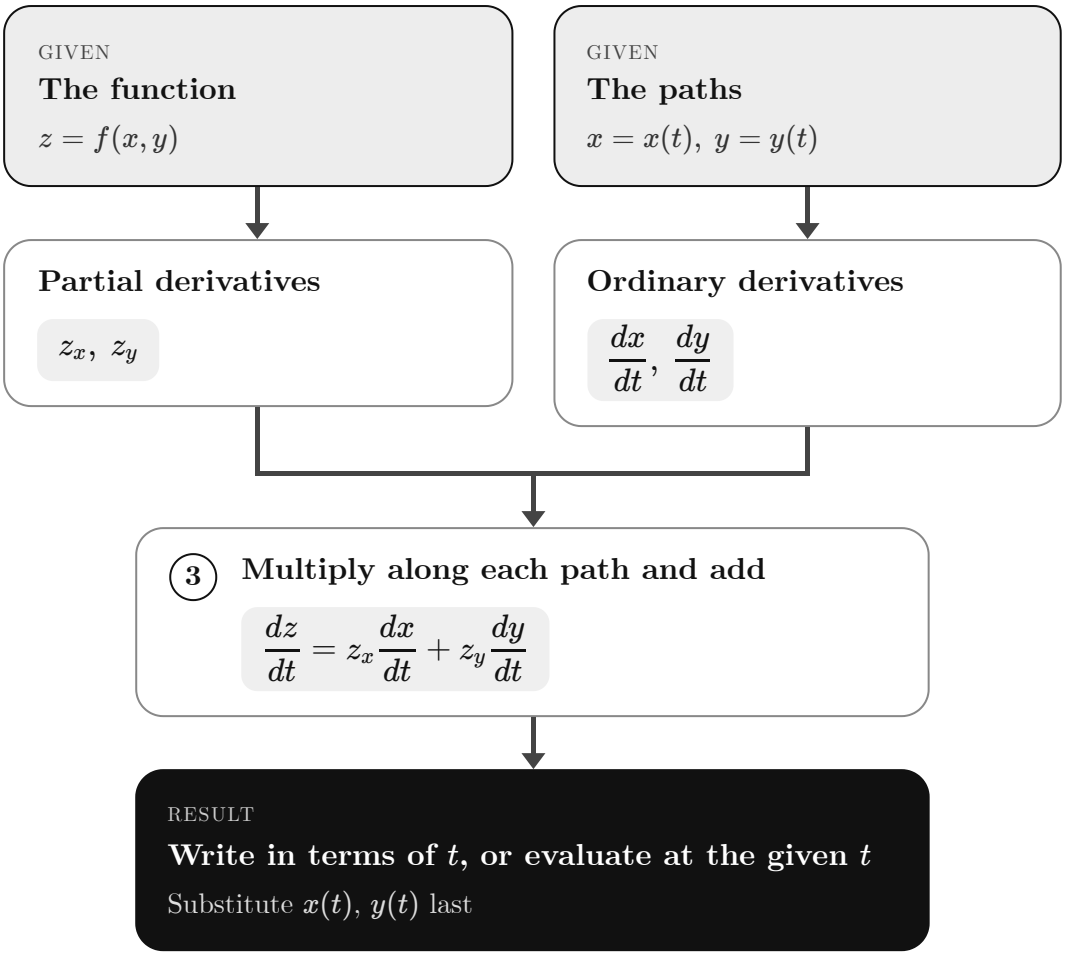


Remark. The multivariable rules in this pack work the same way, except that each intermediate variable adds its own term.

Figure F1.1. Differentiating a composite function from the outside in. ESA ★

S003·4 · T003a 00:00:33

2 The total derivative rule

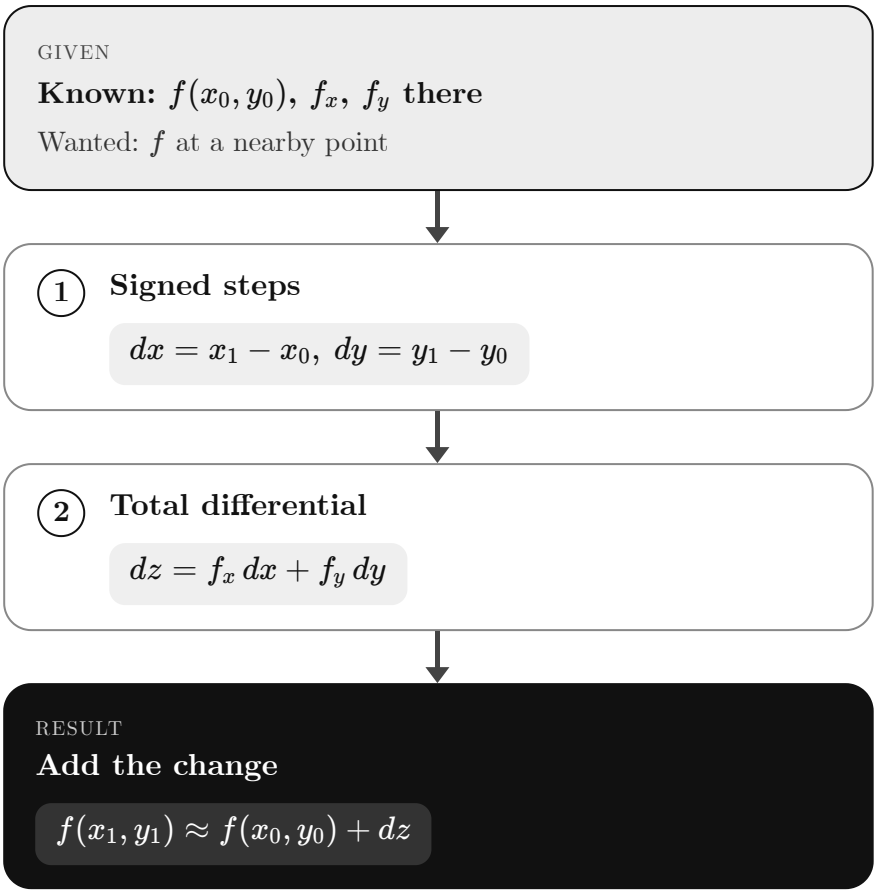


Remark. Check on a small example by substituting first: both routes must agree. On slide 003·6 they do not, which exposes the slide's error (W2.1).

Figure F2.1. Total derivative in three moves. ESA ★★

S003·5 · S003·18 · T003b 00:01:10

3 The total differential: approximation and sensitivity

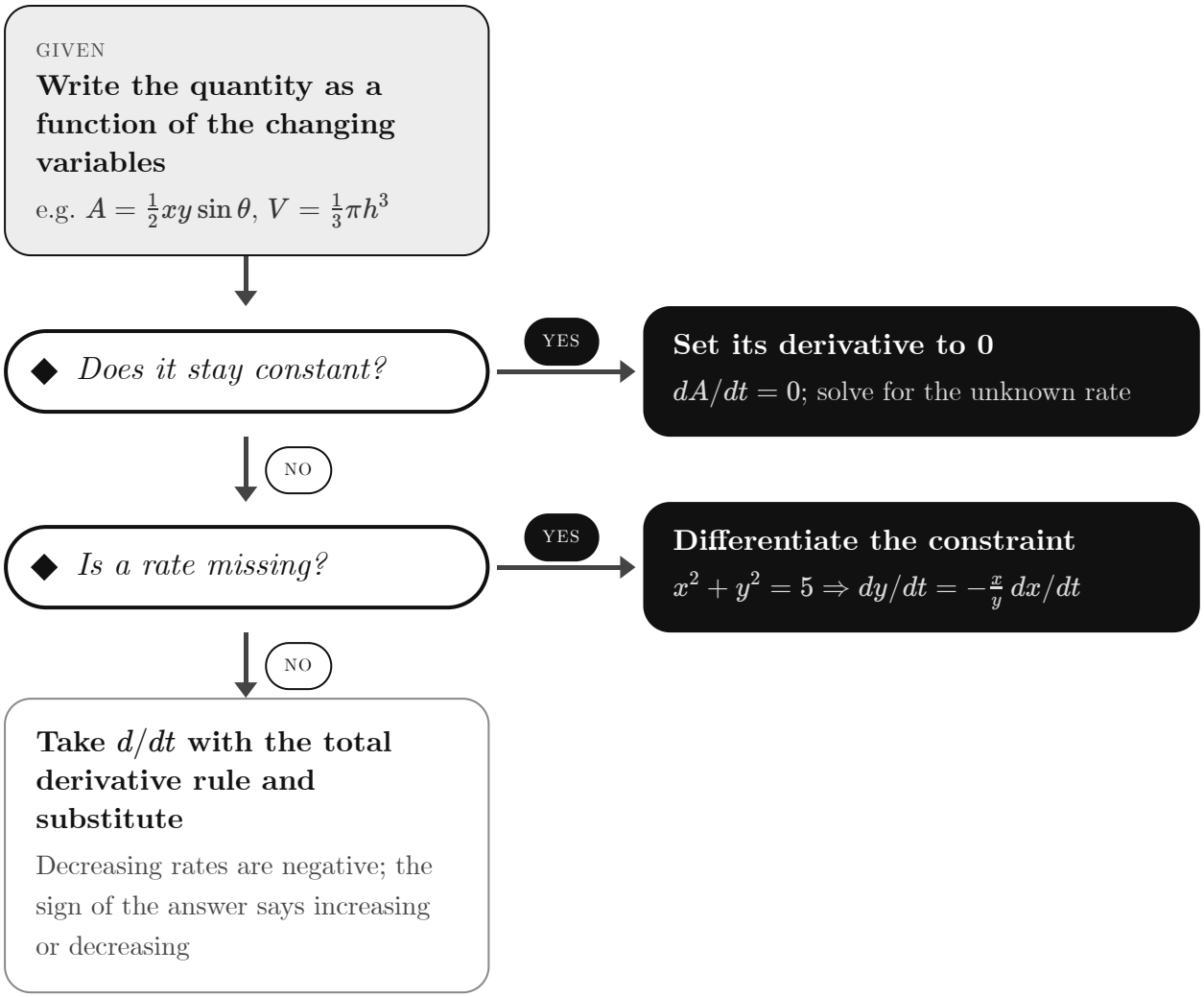


Remark. Slide 003·8: $dz = 1.3(0.1) + (-0.6)(-0.03) = 0.148$, so $f(2.1, -3.03) \approx 6.148$.

Figure F3.1. Estimating a value near a known point. ESA ★★

S003·8 · T003a 00:13:09

4 Rates of change problems

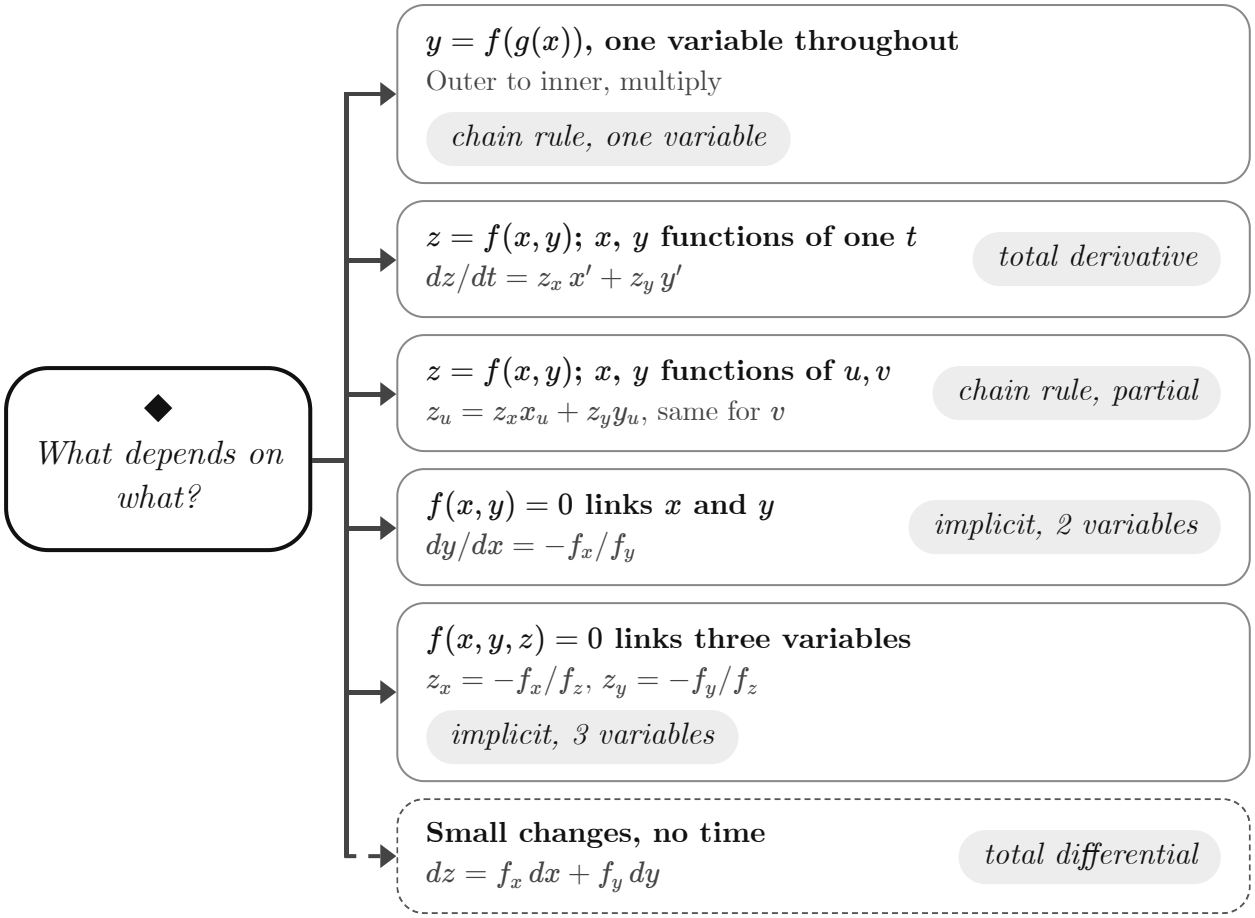


Remark. Both checks can apply in one problem; work down the ladder, then substitute the numbers last.

Figure F4.1. Solving a rates-of-change problem. ESA ★★

S003·7 · S003·9 · S003·21 · T003a 00:07:25

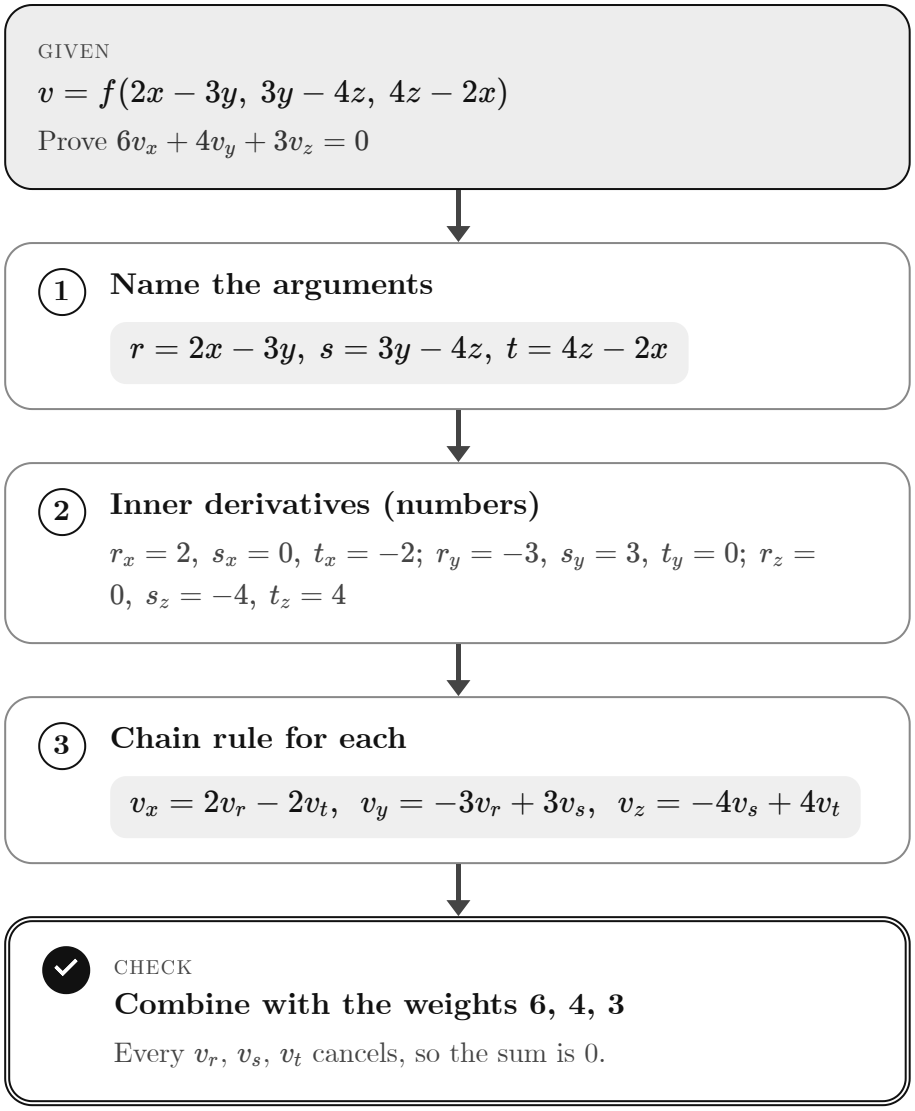
5 The chain rule for composite functions



Remark. The dashed route has no derivative with respect to a variable: it estimates a change. Everything else starts from the tree diagram.

Figure F5.1. Which rule? Read the chain first. ESA ★★★

S003·5 · S003·13 · S004·5 · S004·6 · T004b 00:02:31



Exam trap. Slide 003·16 drops the 3 in $v_y = -3v_r + 3v_s$; without it the terms do not cancel.

Figure F5.2. Proving an identity such as $6v_x + 4v_y + 3v_z = 0$. ESA ★★★

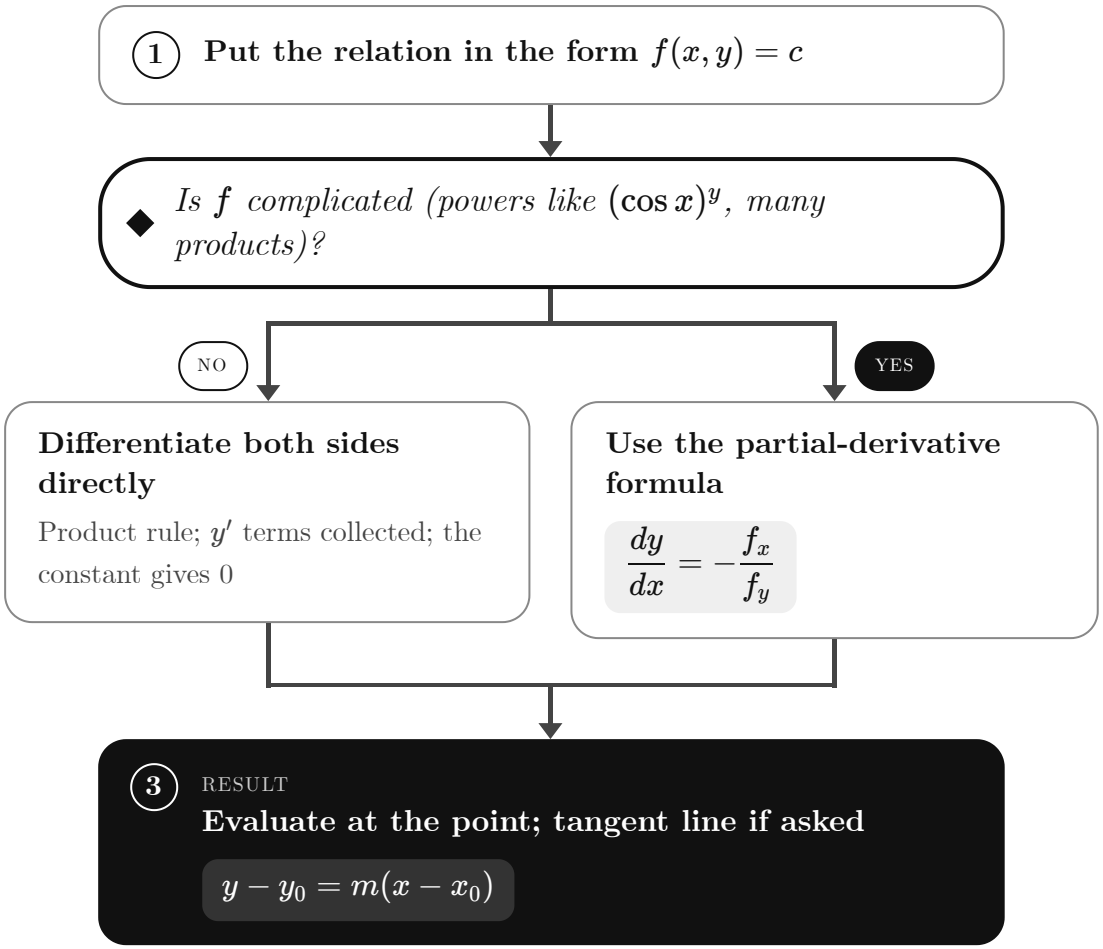
S003·15 · S003·16 · ESA Jan-24 s2 1d

Total derivative vs chain rule vs implicit differentiation

	Total derivative	Chain rule (partial)	Implicit function
Set-up	$z = f(x, y)$, $x(t)$, $y(t)$	$z = f(x, y)$, $x(u, v)$, $y(u, v)$	$f(x, y) = 0$ or $f(x, y, z) = 0$
Final variables	one: t	two: u, v	x (and y)
Formula	$z_x\, x' + z_y\, y'$	$z_x x_u + z_y y_u$	$-f_x/f_y$, $-f_x/f_z$
Result written	dz/dt	$\partial z/\partial u$, $\partial z/\partial v$	dy/dx, $\partial z/\partial x$
Typical question	rate of change at an instant	prove an identity	slope or tangent line

ADDED FOR CLARITY S003·5 · S003·13 · S004·4 · S004·6

6 Implicit functions of two variables

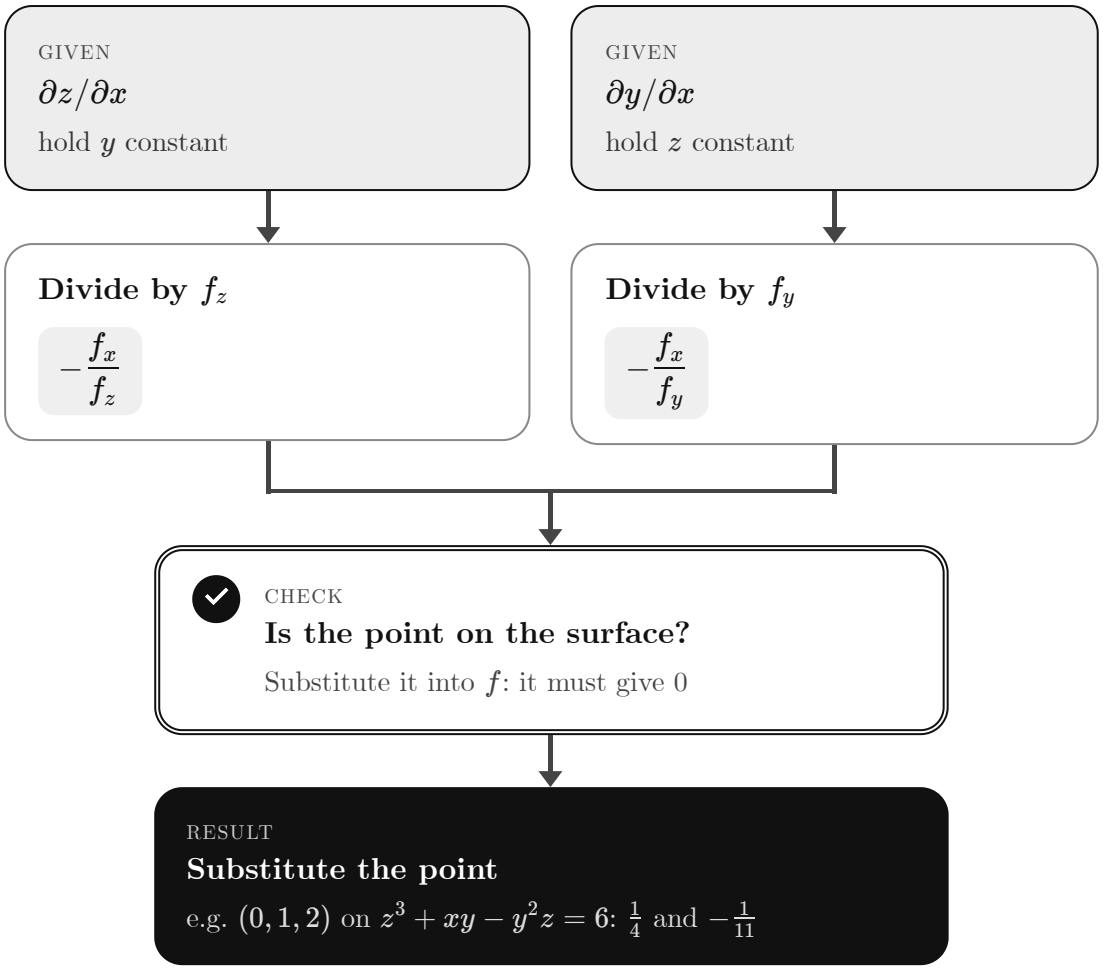


Remark. Both routes give the same dy/dx (W6.1, W6.2). The formula needs $f_y \neq 0$ at the point.

Figure F6.1. Finding dy/dx for an implicit function: two routes. ESA ★★

S003·27 · S004·3 · S004·5 · T003b 00:10:33 · T004a 00:06:19

7 Implicit functions of three variables



Exam trap. The lecture used $(1,0,2)$, which is not on the surface; always check the point first.

Figure F7.1. Partial derivatives from $f(x,y,z)=0$. ESA ★★

S004·6 · S004·9 · S003·26